

Free your Android

Linux Days · Spotlight 4

Fabio Giovanazzi · 19.05.2026



Intro

Link to the slides

Link to the slides



<https://stypox.org/thealt-free-your-android.pdf>

Free software



Types of apps

Basic



Clock



Phone



Gallery



Launcher



PDFs/books/files



Instrument tuner



Local music player



Expense tracker



Weather

Basic+

Basic+Sync



Passwords/OTPs



Calendar



Notes



Flashcards

Basic+Root



Full backups

Complex



Maps/navigation



Browser



Camera



Assistant



Video editor

Online



YouTube



GitHub

Restricted



Banking



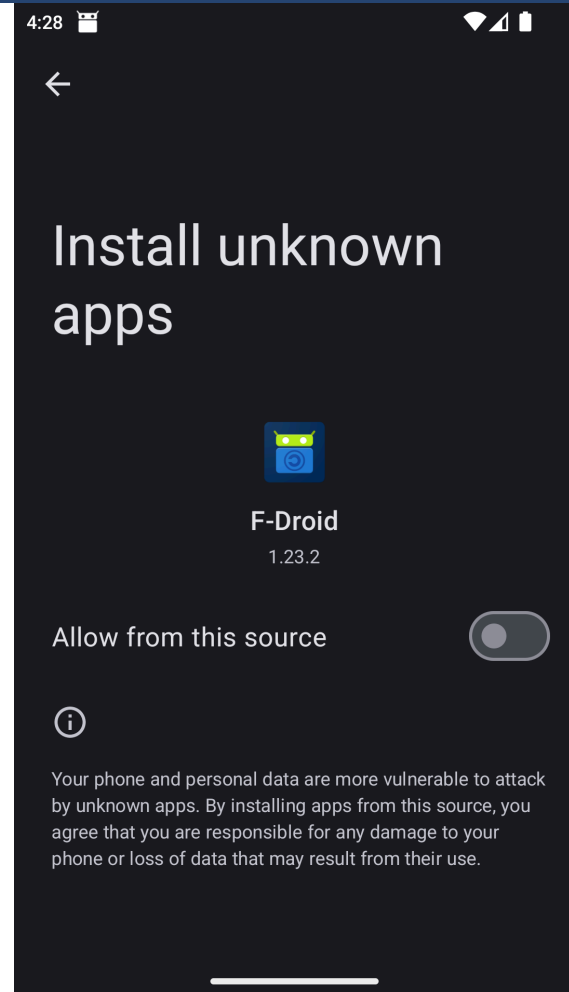
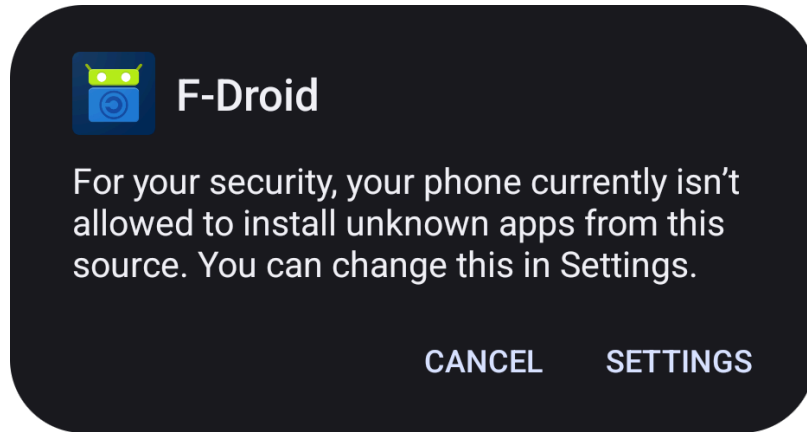
Specific games



WhatsApp

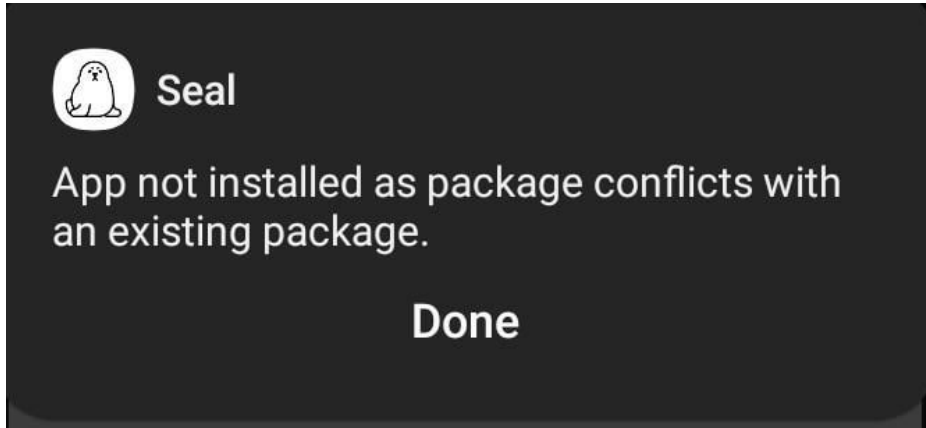
App stores

Enabling unknown sources



Signatures

- APKs must be **cryptographically signed** to be installable on Android
- Signed by developer or by app store



Auto updates

If app previously installed from different store

→ auto-updates not allowed!

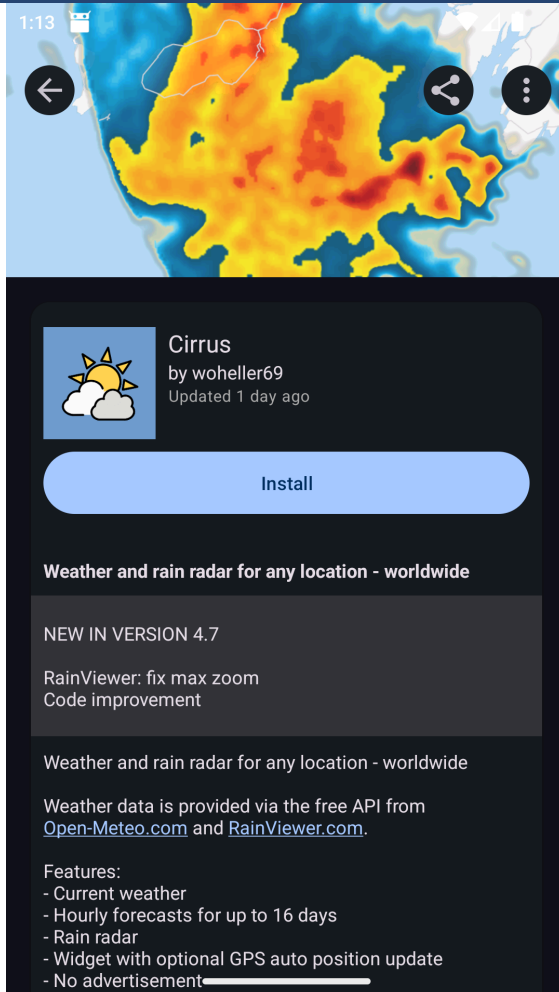
→ need to manually update once

F-Droid

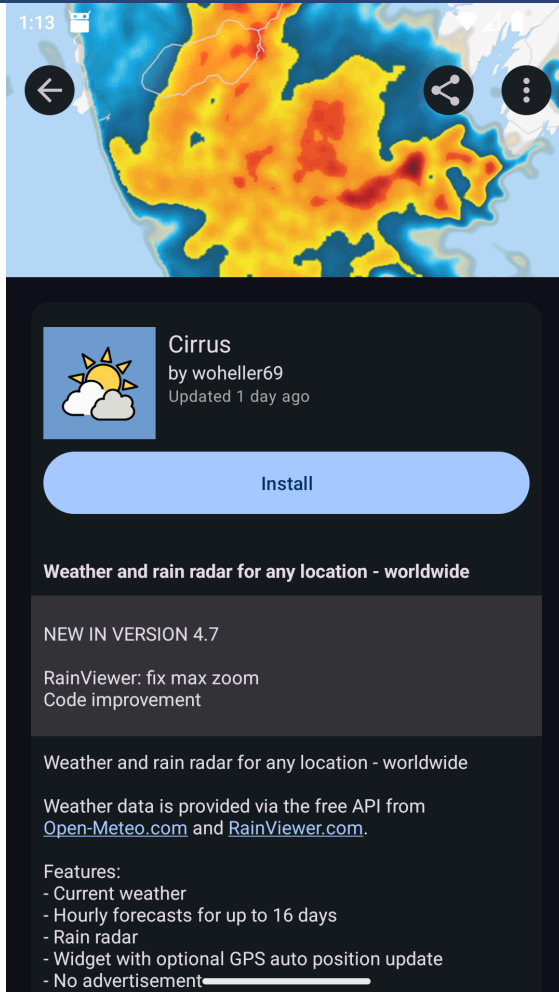


- Cornerstone of libre Android
- **Strict policies** require apps to be fully FOSS
- *Suggestion:* use whenever possible

F-Droid



F-Droid



1:13

Cirrus
by woheller69
Updated 1 day ago

Install

Weather and rain radar for any location - worldwide

NEW IN VERSION 4.7

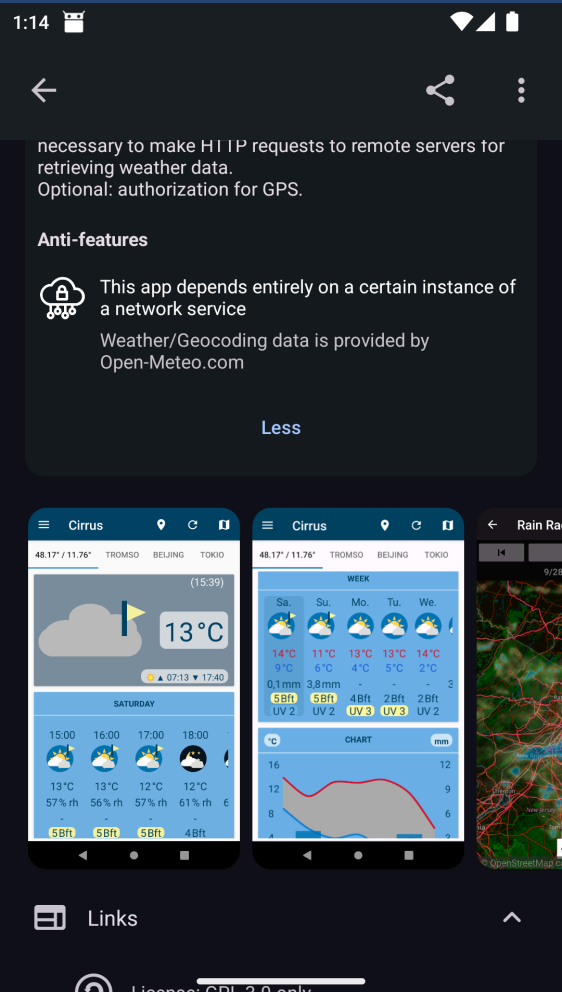
RainViewer: fix max zoom
Code improvement

Weather and rain radar for any location - worldwide

Weather data is provided via the free API from [Open-Meteo.com](https://open-meteo.com) and [RainViewer.com](https://rainviewer.com).

Features:

- Current weather
- Hourly forecasts for up to 16 days
- Rain radar
- Widget with optional GPS auto position update
- No advertisement



1:14

necessary to make HTTP requests to remote servers for retrieving weather data.
Optional: authorization for GPS.

Anti-features

This app depends entirely on a certain instance of a network service
Weather/Geocoding data is provided by Open-Meteo.com

Less

Cirrus

48.17° / 11.76° TROMSO BEIJING TOKIO

(15:39)

13°C

07:13 17:40

SATURDAY

15:00 16:00 17:00 18:00

13°C 13°C 12°C 12°C

57% rh 56% rh 57% rh 61% rh

5Bft 5Bft 5Bft 4Bft

WEEK

Sa Su Mo Tu We

14°C 11°C 13°C 13°C 14°C

9°C 6°C 4°C 5°C 2°C

0.1mm 3.8mm - - -

5Bft 5Bft 4Bft 2Bft 2Bft

UV 2 UV 2 UV 3 UV 3 UV 2

CHART

°C mm

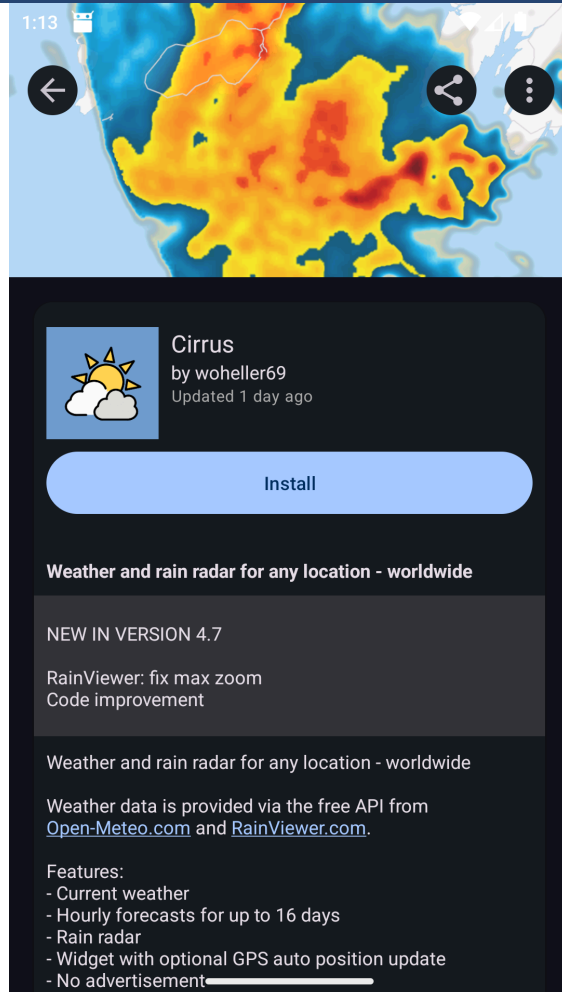
16 12 8 4

15 12 9 6

Links

License: GPL-3.0-only

F-Droid



1:13

Cirrus
by woheller69
Updated 1 day ago

Install

Weather and rain radar for any location - worldwide

NEW IN VERSION 4.7

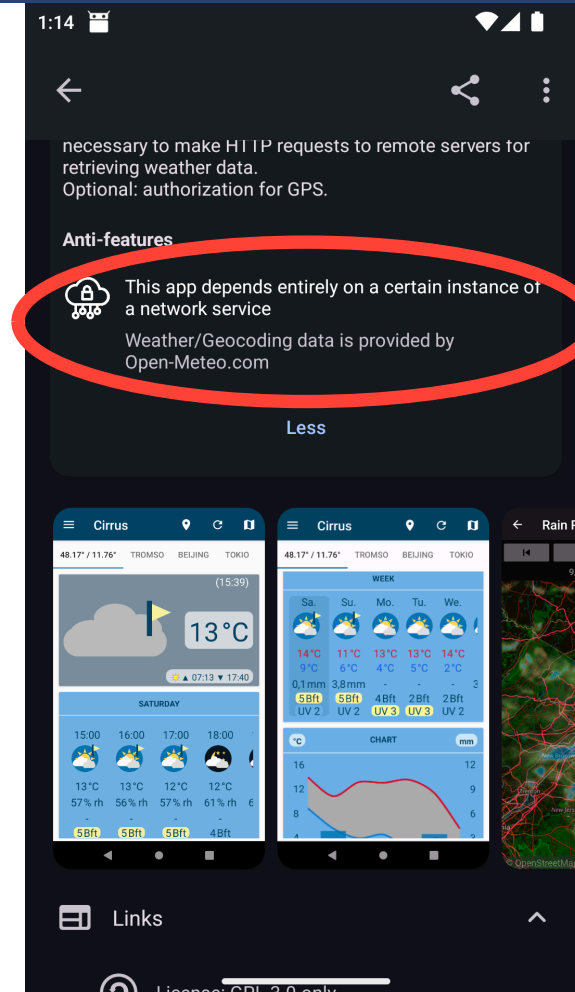
RainViewer: fix max zoom
Code improvement

Weather and rain radar for any location - worldwide

Weather data is provided via the free API from [Open-Meteo.com](https://open-meteo.com) and [RainViewer.com](https://rainviewer.com).

Features:

- Current weather
- Hourly forecasts for up to 16 days
- Rain radar
- Widget with optional GPS auto position update
- No advertisement



1:14

necessary to make HTTP requests to remote servers for retrieving weather data.
Optional: authorization for GPS.

Anti-features

 This app depends entirely on a certain instance of a network service
Weather/Geocoding data is provided by Open-Meteo.com

Less

Cirrus

48.17° / 11.76° TROMSØ BEIJING TOKIO

(15:39)

13°C

07:13 17:40

SATURDAY

15:00 16:00 17:00 18:00

13°C 13°C 12°C 12°C

57% rh 56% rh 57% rh 61% rh

5Bft 5Bft 5Bft 4Bft

WEEK

Sa Su Mo Tu We

14°C 11°C 13°C 13°C 14°C

9°C 6°C 4°C 5°C 2°C

0.1mm 3.8mm - - -

5Bft 5Bft 4Bft 2Bft 2Bft

UV 2 UV 2 UV 3 UV 3 UV 2

CHART

°C mm

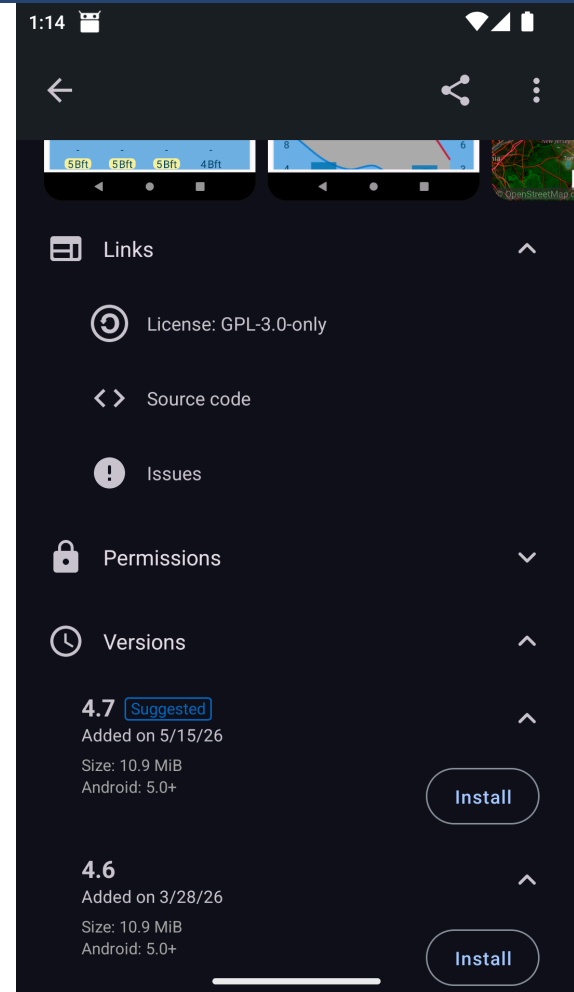
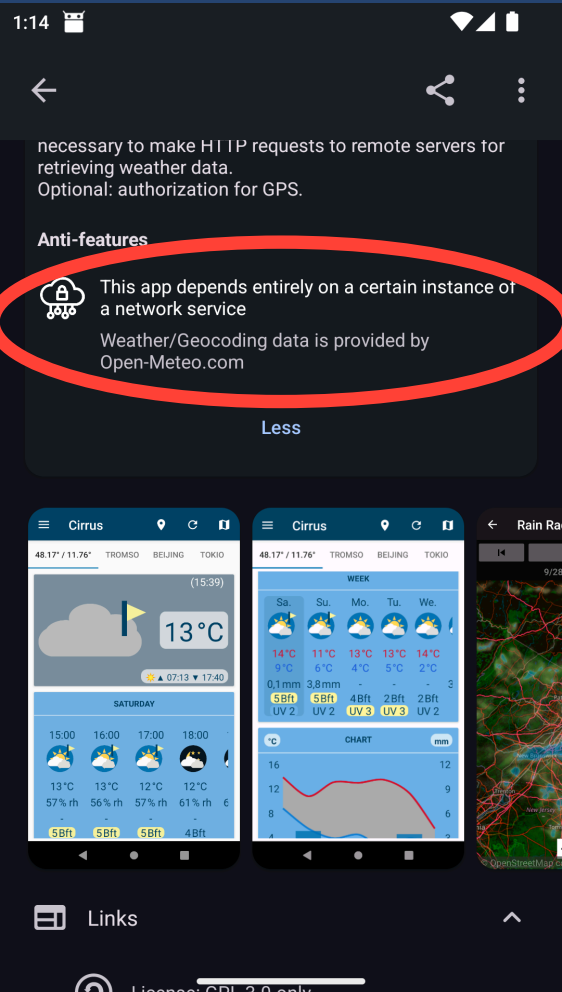
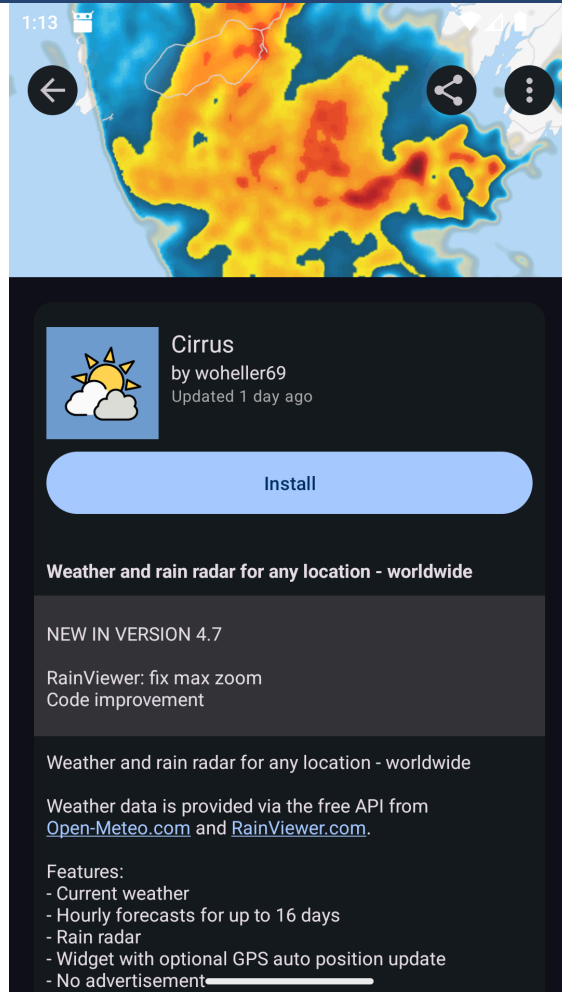
16 12 8 4

12 9 6 3

Links

License: GPL-3.0-only

F-Droid

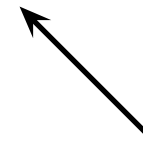


Build process

License: GPL-3.0-only

AntiFeatures: { TetheredNet: { en-US: Weather/Geocoding data is provided by Open-Meteo.com }}

Repo: <https://github.com/woheller69/omweather.git>



Metadata file

Build process

- APKs built by F-Droid's servers

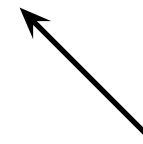
License: GPL-3.0-only

AntiFeatures: { TetheredNet: { en-US: Weather/Geocoding data is provided by Open-Meteo.com }}

Repo: <https://github.com/woheller69/omweather.git>

Builds:

```
- versionName: '1.0'  
  versionCode: 10  
  commit: V1.0  
  gradle: [yes]
```



Metadata file

Build process

- APKs built by F-Droid's servers
- Result checked against developer's APK
 - if successful: **reproducible builds** shipped
 - otherwise: APK signed by F-Droid

License: GPL-3.0-only

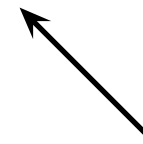
AntiFeatures: { TetheredNet: { en-US: Weather/Geocoding data is provided by Open-Meteo.com }}

Repo: <https://github.com/woheller69/omweather.git>

AllowedAPKSigningKeys: 8fa634071b0fe0fe0832fcc159aa41222448...

Builds:

- versionName: '1.0'
- versionCode: 10
- commit: V1.0
- gradle: [yes]



Metadata file

F-Droid repositories

IzzyOnDroid F-Droid Repository

This is the F-Droid Repository by *IzzyOnDroid* ([details](#)), currently serving **1369** apps. To add it to your F-Droid client, scan the QR code (click it to enlarge) or use this URL:

<https://apt.izzysoft.de/fdroid/repo>

Select repo:

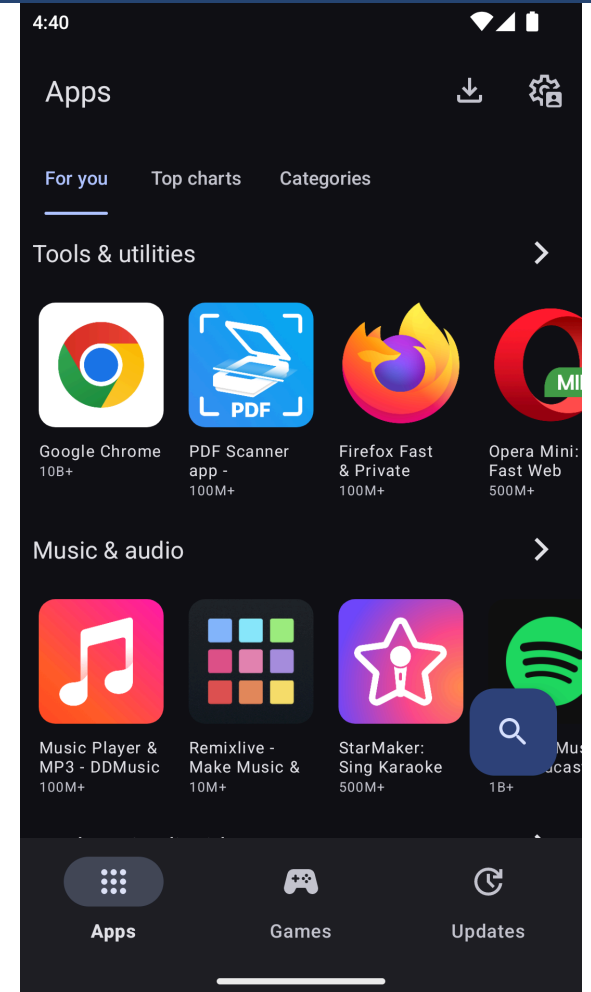
Last updated: 2026-05-15 17:54 UTC



Aurora Store



- Client for Google Play Store
- Login or use anonymous account
- *Suggestion:* use for proprietary apps
- <https://f-droid.org/en/packages/com.aurora.store/>



Accrescent



- **Security**-focused app store
- Apps are signed by original developers
- <https://accrescent.app/>

Obtainium



- Installs apps directly from their **release page**
- f-droid.org/en/packages/dev.imranr.obtainium.fdroid/

App suggestions

Basic apps

- Many good alternatives **on F-Droid**
- Look at crowdsourced lists of FOSS apps, e.g. [this](#)

Basic apps

- Many good alternatives **on F-Droid**
- Look at crowdsourced lists of FOSS apps, e.g. [this](#)
- *Suggestion:* [Fossify](#) suite for streamlined experience

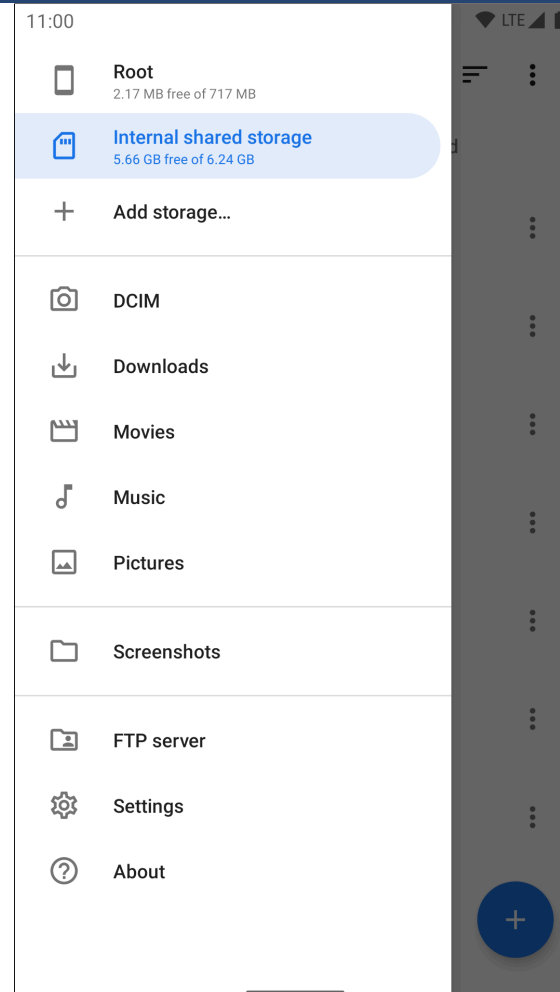


-  Fossify Phone
Unlock a secure, open-source
-  Fossify Paint
Quick, easy and open source
-  Fossify Notes
To do list widget with a noteb
-  Fossify Launcher
Customize your home screen
-  Fossify Calculator
Simple and open-source calcul
-  Fossify Keyboard
Simple and open-source offlin
-  Fossify Camera
Lightweight open-source cam
-  Fossify Calendar
Securely plan, schedule, and s
-  Fossify Clock
Handy, lightweight, open-sour
-  Fossify Voice Recorder
Record anything with this Ope
-  Fossify File Manager
Handy, lightweight, open-sour
-  Fossify Gallery
Gallery with Photo editor. No
-  Fossify Messages
Open-source and Ad-free SMS
-  Fossify Music Player
Open-source and Ad-free Mus
-  Fossify Contacts
Manage your contacts private

Basic apps

Basic - my top picks

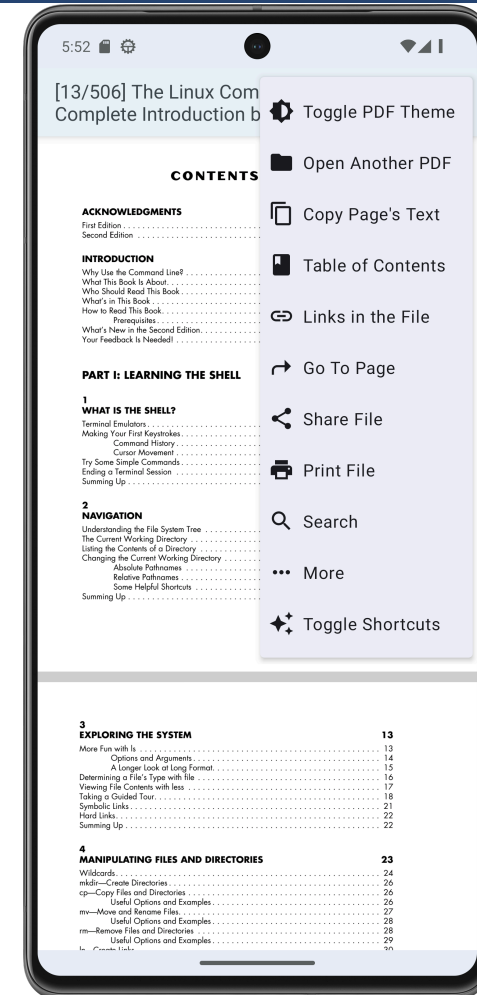
- File manager: [Material Files](#)



Basic apps

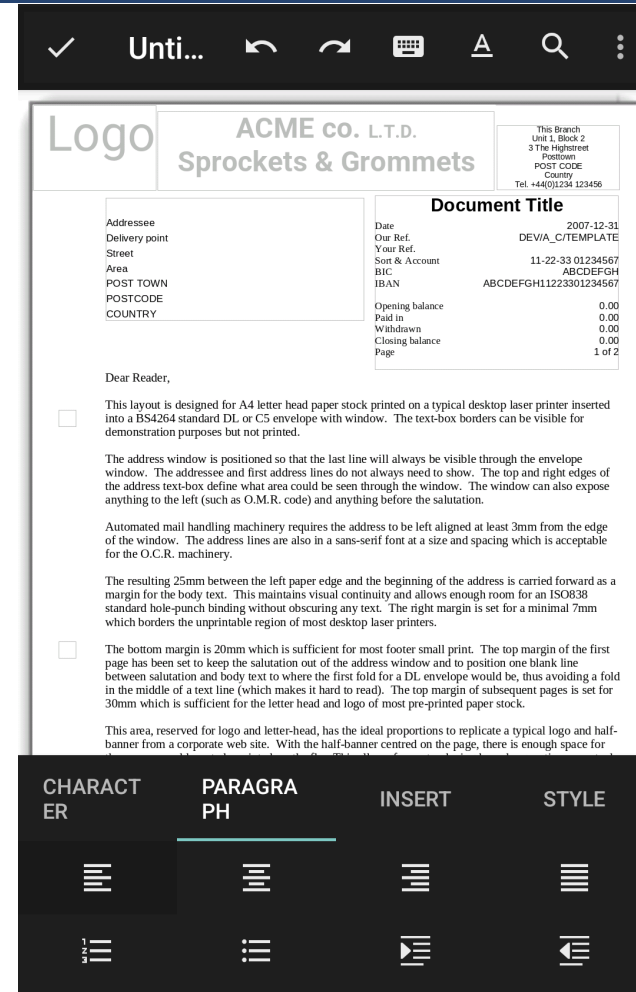
Basic - my top picks

- File manager: [Material Files](#)
- PDF viewer: [MJ PDF](#)



Basic - my top picks

- File manager: [Material Files](#)
- PDF viewer: [MJ PDF](#)
- [LibreOffice Viewer](#)



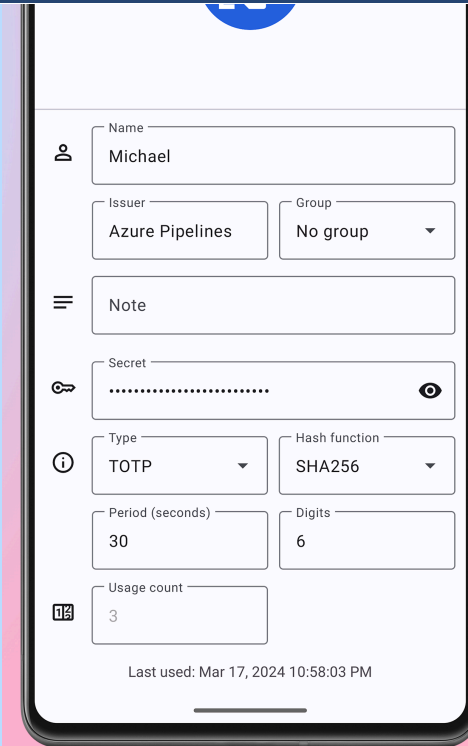
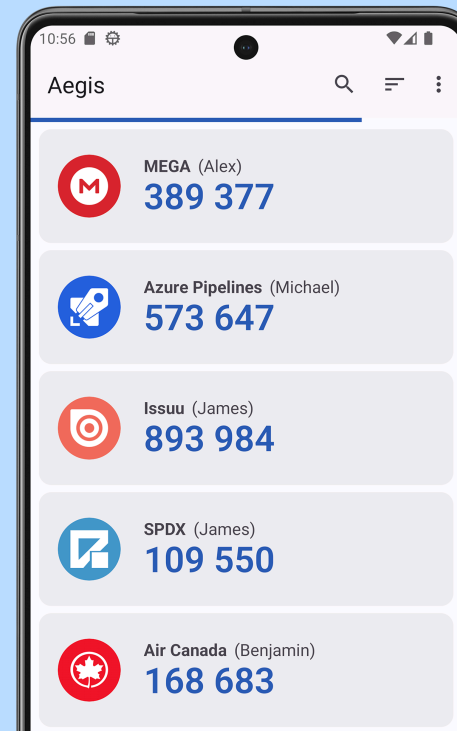
Basic apps

Basic - my top picks

- File manager: [Material Files](#)
- PDF viewer: [MJ PDF](#)
- [LibreOffice Viewer](#)
- OTP: [Aegis](#)

Organized

All your tokens in a polished interface



Expert

Edit your tokens in our advanced editing screen

Syncing across devices

This is **harder**

- May be service-specific
- May require self-hosting
- May require syncing files manually

Basic apps

Syncthing

- Syncs files across devices
- No need to self-host anything
- Can't sync private app data :-(
- <https://syncthing.net/>

The screenshot displays the Syncthing web interface in a browser window. The address bar shows 'localhost' and the URL 'kvin.kastelo.net'. The interface is in English and includes a 'Help' link and an 'Actions' menu.

Folders

Folder	Status
documents	Up to Date
foto	Up to Date
Folder Details for 'foto':	
Folder ID	gw4hx-3dz2m
Folder Path	/Volumes/foto
Global State	104,194 files, 587 folders, ~997 GiB
Local State	104,193 files, 587 folders, ~997 GiB
Folder Type	Send Only
Ignore Permissions	Yes
Rescans	1d, Disabled
Shared With	pseudo.nym.se
Last Scan	2020-05-18 12:14:19
Reduced by ignore patterns	
Buttons: Pause, Rescan, Edit	
foto/2020	Up to Date
foto/albums	Up to Date
kastelo	Up to Date
lightroom	Up to Date
Buttons: Pause All, Rescan All, Add Folder	

This Device

Device	Status
kvin.kastelo.net	Up to Date
Device Statistics:	
Download Rate	0 B/s (928 B)
Upload Rate	0 B/s (980 B)
Local State (Total)	112,059 files, 1,291 folders, ~1.01 TiB
Listeners	2/2
Discovery	4/5
Uptime	1m
Version	v1.6.0-rc.4+22-g5b34c31c, macOS (64 bit)

Remote Devices

Device	Status
dso	Up to Date
kvar.kastelo.net	Disconnected
pseudo.nym.se	Up to Date
Device Statistics for pseudo.nym.se:	
Download Rate	0 B/s (640 B)
Upload Rate	0 B/s (595 B)
Address	172.16.32.11:22000
Version	v1.6.0-rc.4
Folders	documents, foto, foto/albums, kastelo, lightroom
Buttons: Pause, Edit	
uploads	Paused (Unused)
Buttons: Pause All, Resume All, Recent Changes, Add Remote Device	

Footer

Home page | Documentation | Support | Statistics | Changelog | Bugs | Source Code | Twitter

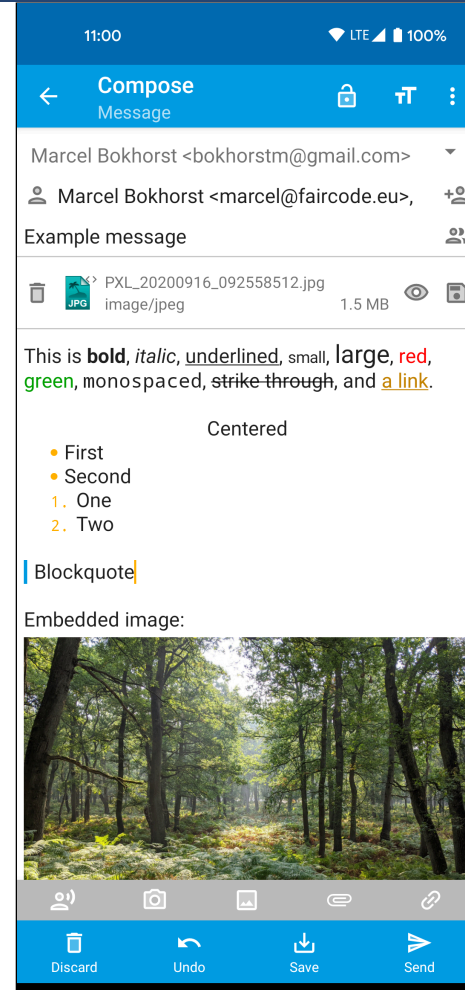
Cloud

- Self-host or buy a subscription
- Comes with synced tools: notes, passwords, calendar, ...
- [NextCloud](#)
- [OwnCloud](#)
- [OpenCloud](#)
- ...

Basic apps

Basic+Sync - my top picks

- Email: [FairEmail](#)



Basic apps

Basic+Sync - my top picks

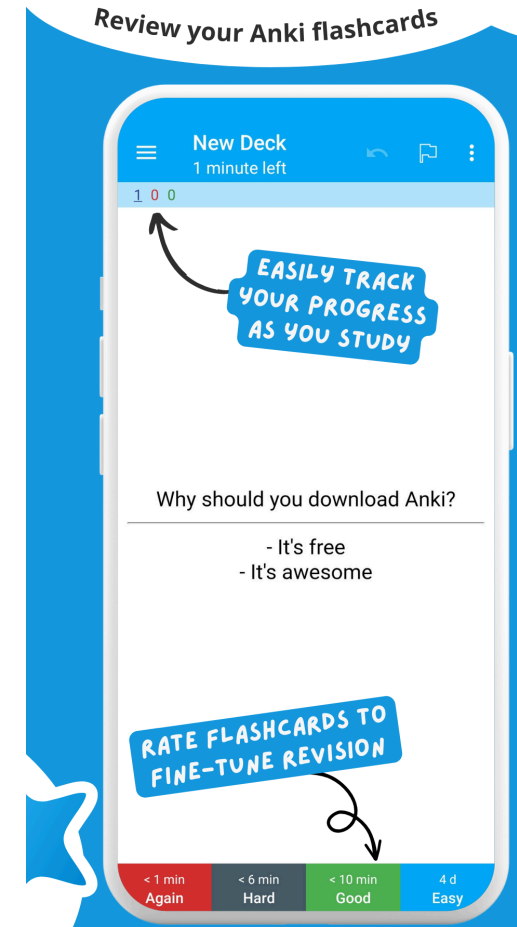
- Email: [FairEmail](#)
- Synced calendar: [Etar](#), [Fossify Calendar](#)



Basic apps

Basic+Sync - my top picks

- Email: [FairEmail](#)
- Synced calendar: [Etar](#), [Fossify Calendar](#)
- Synced flashcards: [Anki](#)



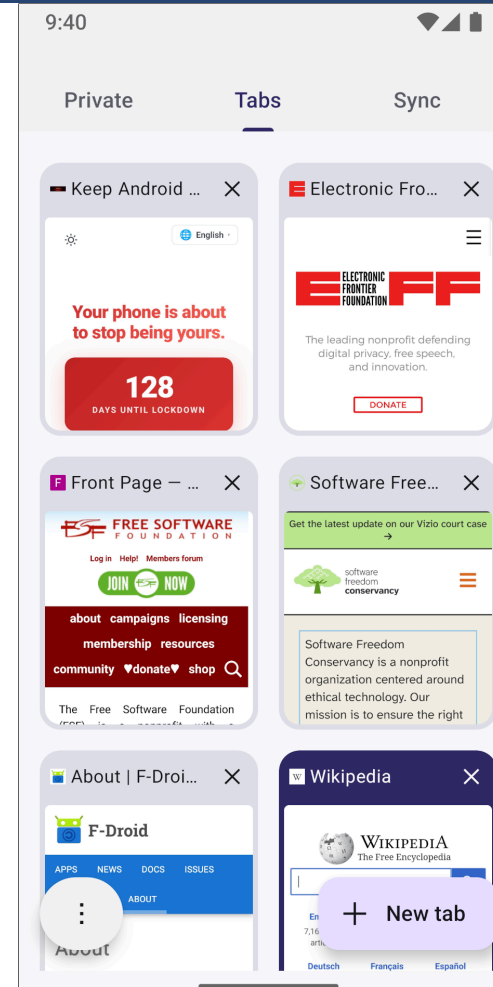
Complex apps

- complicated functionality
 - opinionated feature set
- difficult to find **exact replacements**

Complex apps

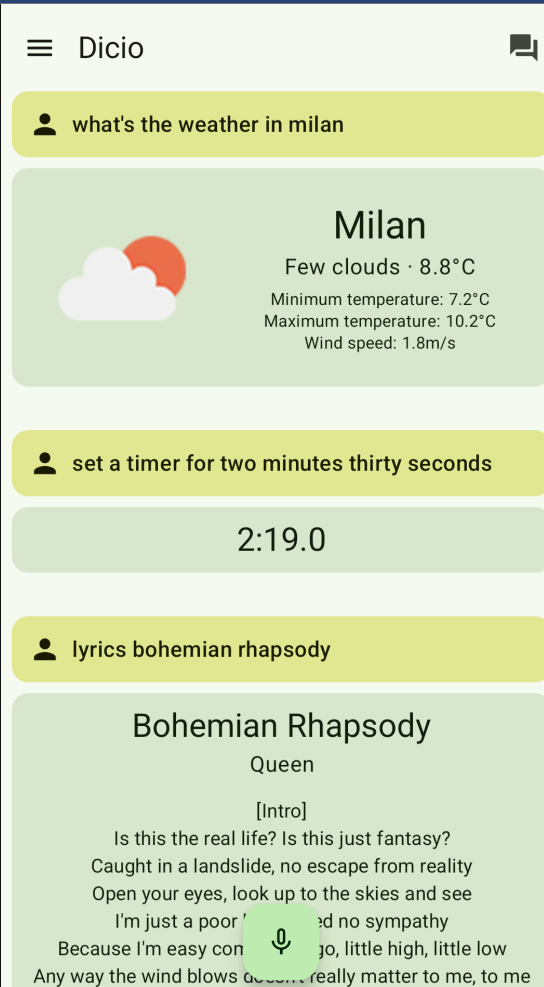
Examples

- Browser: Firefox/[Fennec](#) → plugin support!



Examples

- Browser: Firefox/[Fennec](#) → plugin support!
- Assistant: [Dicio](#), but not on par



Examples

- Browser: Firefox/[Fennec](#) → plugin support!
- Assistant: [Dicio](#), but not on par
- Camera: none are comprehensive :-(
 - Open Camera → featureful but clunky UI
 - Fossify Camera → limited

Maps

Great options!

- offline
- based on OpenStreetMap data → good for hikes
- public transport support may be lacking

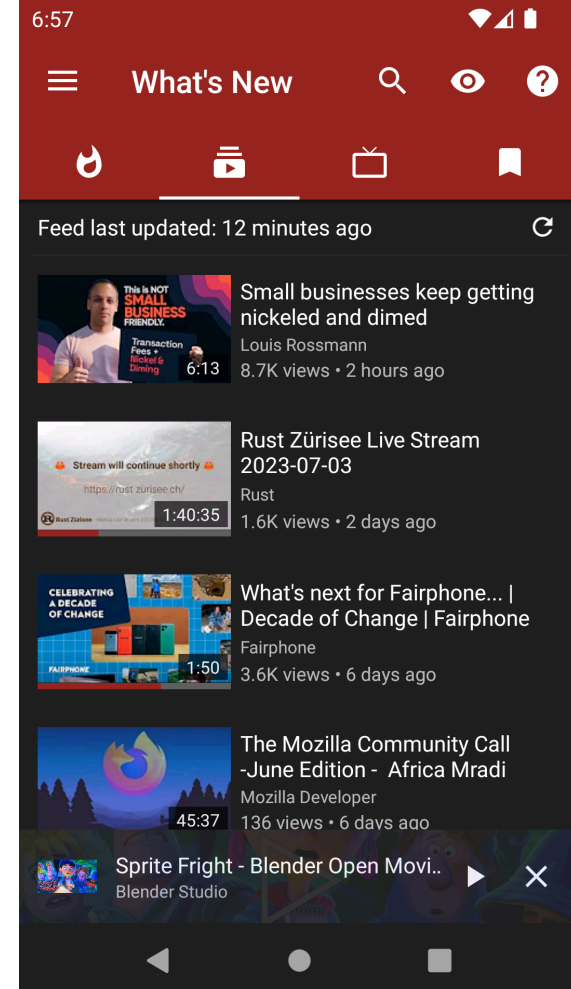
Top picks

- [OsmAnd~](#)
- [Organic Maps](#) / [CoMaps](#)
- [Transportr](#)

Online apps

Featureful alternative clients can replace official apps:

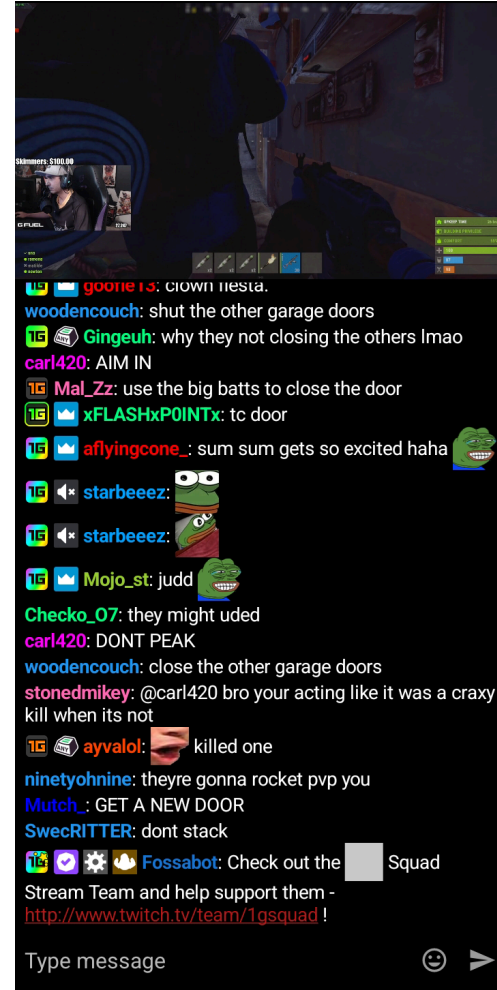
- YouTube (Music): [NewPipe](#), [LibreTube](#), ...



Online apps

Featureful alternative clients can replace official apps:

- YouTube (Music): [NewPipe](#), [LibreTube](#), ...
- Twitch: [Twire](#)



Online apps

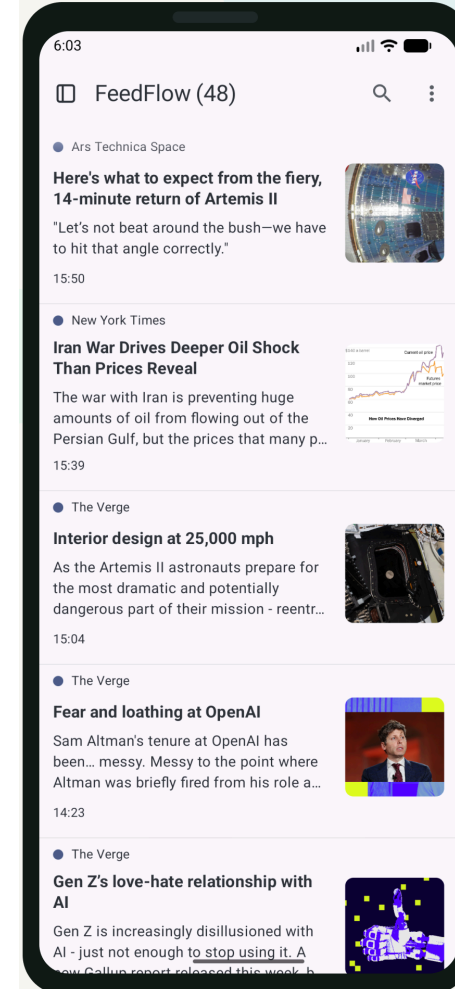
Featureful alternative clients can replace official apps:

- YouTube (Music): [NewPipe](#), [LibreTube](#), ...
- Twitch: [Twire](#)
- Telegram: [Forkgram](#)
- ...

Online apps

Featureful alternative clients can replace official apps:

- YouTube (Music): [NewPipe](#), [LibreTube](#), ...
- Twitch: [Twire](#)
- Telegram: [Forkgram](#)
- ...
- [FeedFlow](#) for RSS



Non replaceable apps

Some apps can't be replaced :-)

- Banking apps
- Instagram
- Specific games
 - but F-Droid has a nice collection of FOSS games
 - e.g. [Shattered Pixel Dungeon](#)

Other gems

- [MensaZH](#)



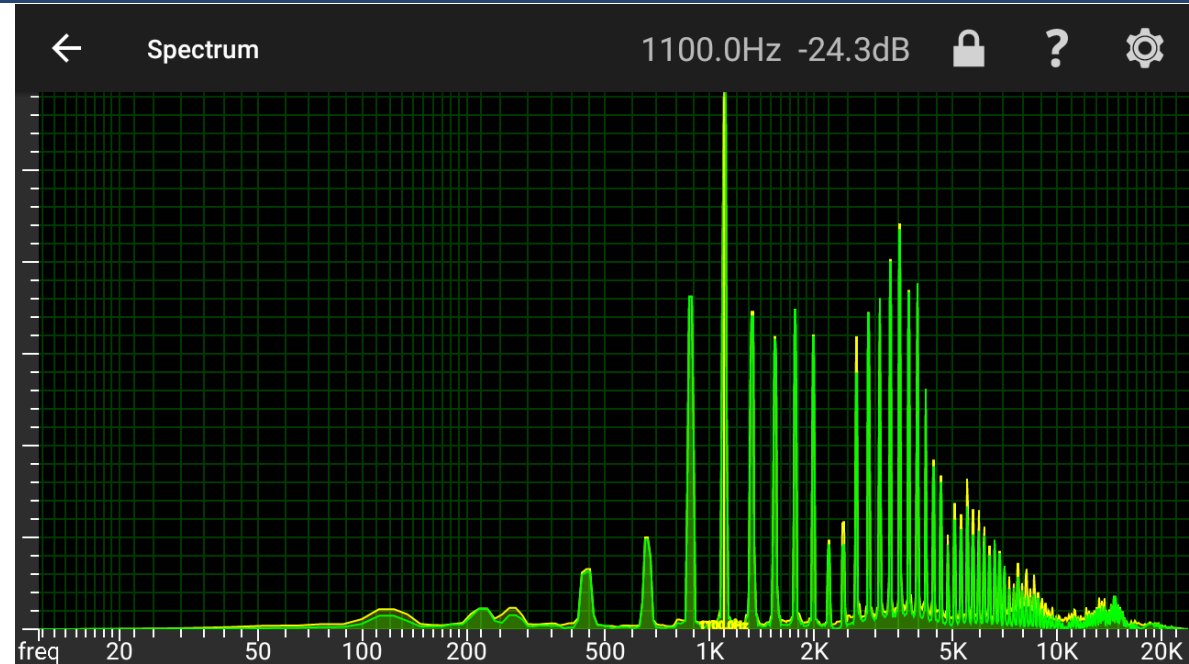
Other gems

- [MensaZH](#)
- [PeakOrama](#), [Sky Map](#)



Other gems

- [MensaZH](#)
- [PeakOrama](#), [Sky Map](#)
- [Oscilloscope](#)



Other gems

Other gems

- [MensaZH](#)
- [PeakOrama](#), [Sky Map](#)
- [Oscilloscope](#)
- [Tabs Lite](#) for chords

← 🔍 ❤️ ➦ ⋮

Mamma Mia by [ABBA](#)

Difficulty: novice
Tuning: E A D G B E
Capo: None
Key: D
Tab Author: Unregistered
Transpose: 0 ✕ — +
Version 2 ▾

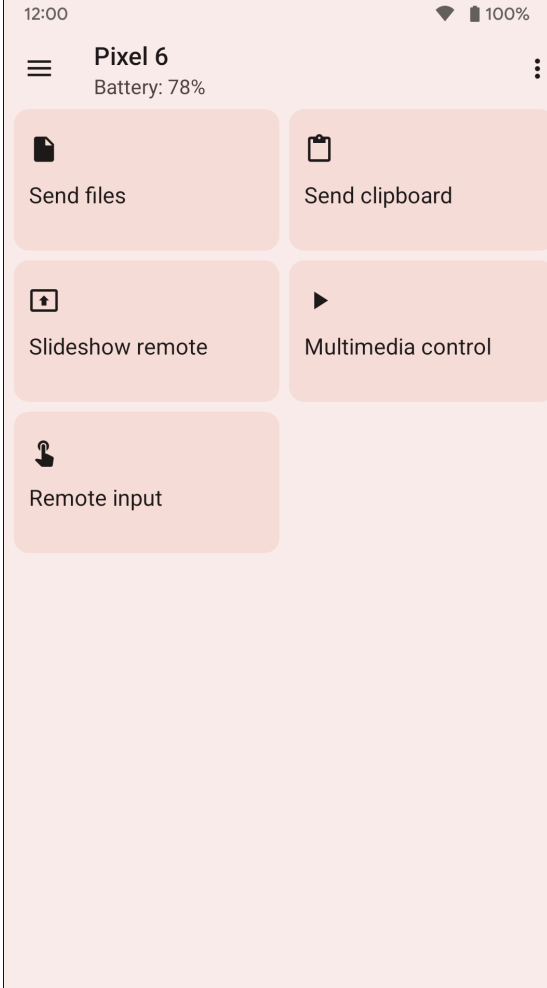
[Verse 1]
D G
I've been cheated by you since I don't know when
D G
So I made up my mind, it must come to an end
D5 D+5
Look at me now, will I ever learn?
D6 D7 G
I don't know how but I suddenly lose control
G A
There's a fire within my soul

[Pre-Chorus]
G D A
Just one look and I can hear a bell ring
G D A
One more look and I forget everything, o-o-o-oh

[Chorus]

Other gems

- [MensaZH](#)
- [PeakOrama](#), [Sky Map](#)
- [Oscilloscope](#)
- [Tabs Lite](#) for chords
- [KDE Connect](#)



Degoogling

Disabling system apps

... even if the Disable button is grayed out

Manually (disabling)

Find the package name of the app to disable:

- System settings → Apps → the app to inspect → scroll to bottom
- Or find out with a system manager

Disable it ("**app.package.name**" is the package name):

- `adb shell cmd package disable-user --user 0 "app.package.name"`

You may re-enable it like this:

- `adb shell cmd package enable --user 0 "app.package.name"`

Manually (uninstalling)

Sometimes disabling may not work

→ system apps not globally uninstallable

→ but uninstallable for specific user!

Uninstall:

- `adb shell cmd package uninstall -k --user 0 "app.package.name"`

Reinstall (only if system app still installed globally):

- `# install-existing always installs for the current user, i.e. usually user 0`
`adb shell cmd package install-existing "app.package.name"`

With a debloating script

People have compiled lists of apps that can be disabled safely

Google Play Services

- Proprietary background services and libraries
- Many apps require them :-(

MicroG

- FOSS **reimplementation** of Google Play Services
- Installed as system app
- Still contacts Google servers
- [Important APIs work well](#)

microG Settings

Self-Check
Check if the system is correctly set up to use microG.

Google Services

Google Accounts
Add and manage Google accounts

Google device registration
On

Cloud Messaging
On - 7 registered Apps

Google SafetyNet
Off

Play Store services

Work profile

Location

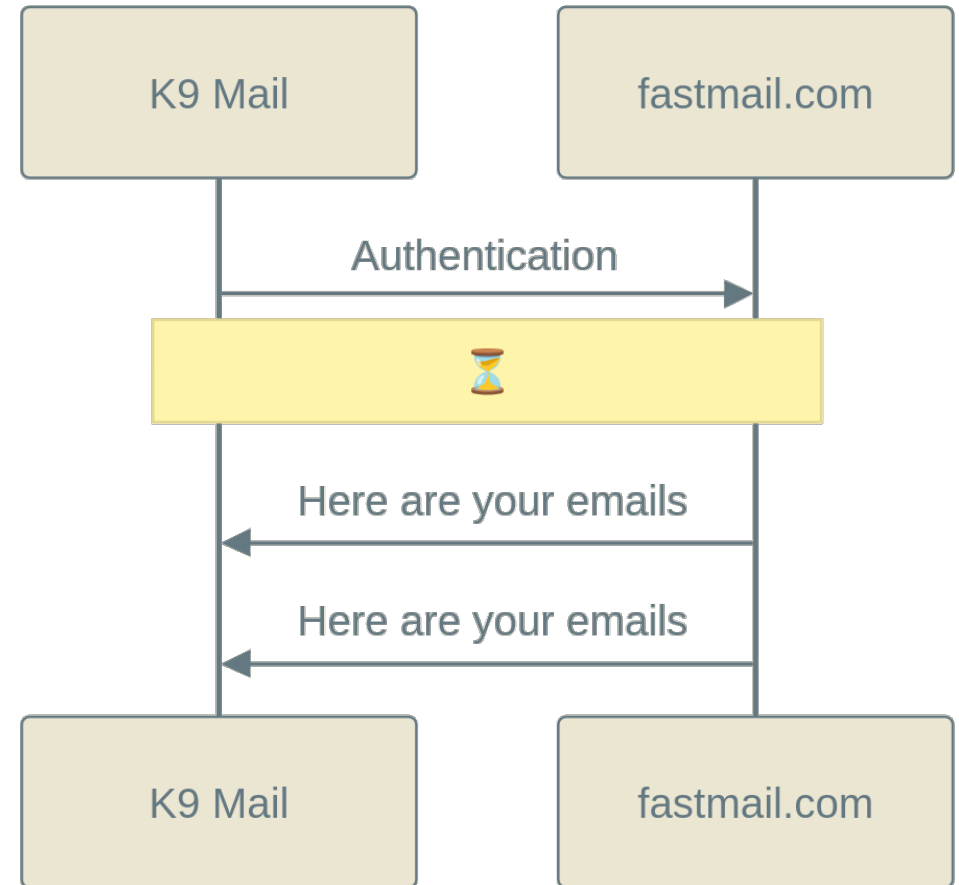
About microG Services
Version 0.3.11.250932

Android Auto

- Android Auto requires proprietary app as system app :-/
- Some ROMs preinstall stubs

Receiving notifications by polling

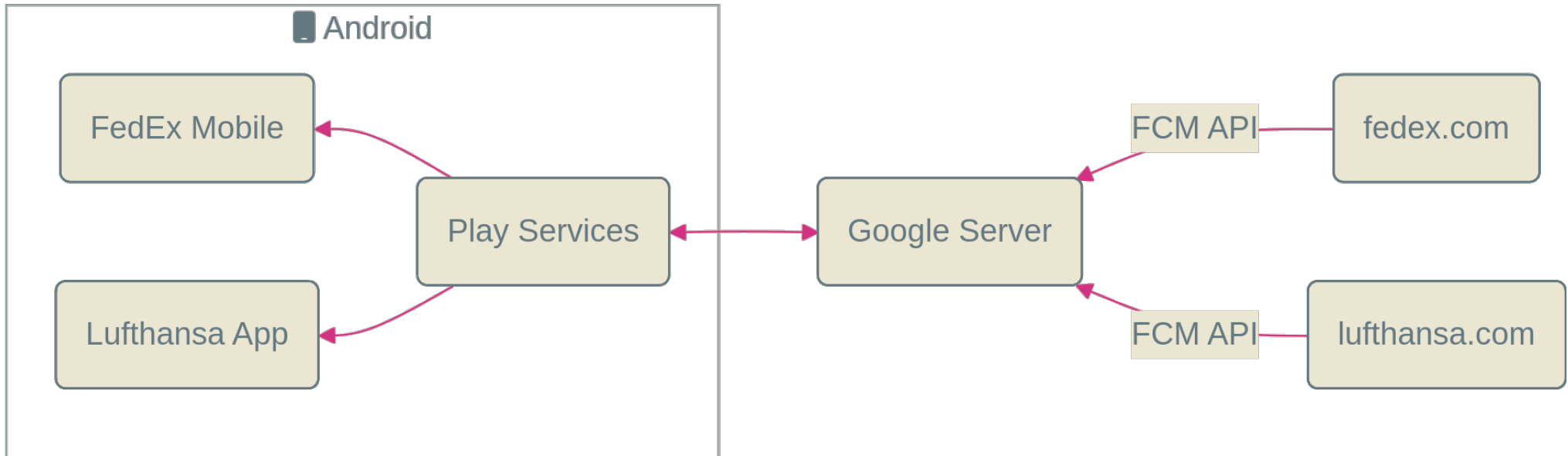
- Every app keeps a connection open
- Battery intensive



[Source](#), © Simon Gougeon, Daniel Gultsch

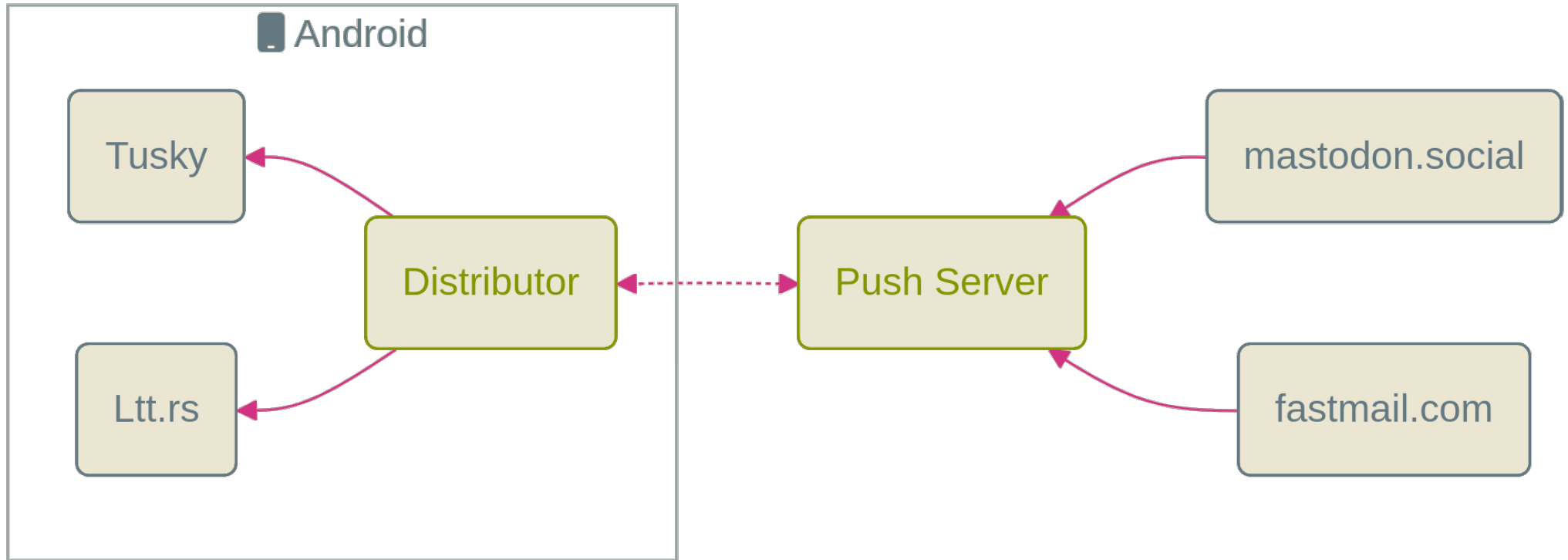
Firebase Cloud Messaging

Only **one** connection open for **all** apps



For more info watch [this FOSDEM 2026 presentation](#), © Simon Gougeon, Daniel Gultsch

UnifiedPush



For more info watch [this FOSDEM 2026 presentation](#), © Simon Gougeon, Daniel Gultsch

Attestation

Can attest authenticity of hardware, firmware, OS and app

1. Ensure that no attacker has tampered with the phone
2. Server can trust computations of a remote device

See [GrapheneOS' docs](#).

Hardware attestation

- Based on Trusted Execution Environment or Hardware Security Module
- Certificates chain up to Google root certificate
- Apps can whitelist keys of custom ROMs

Play Integrity API (SafetyNet)

- Uses DroidGuard
- May not be backed by hardware attestation
- Relies on Google's servers to provide a challenge
- **Custom ROMs can't implement it**

Android ROMs

Android Open Source Project (AOSP)

- Android source code released twice a year
- Google provides monthly security patches
- Does not contain devices' HAL

LineageOS (ex CyanogenMod)

- **Up-to-date** Android on **old phones**
- Supports 279 devices
- Live demo later!
- <https://lineageos.org/>



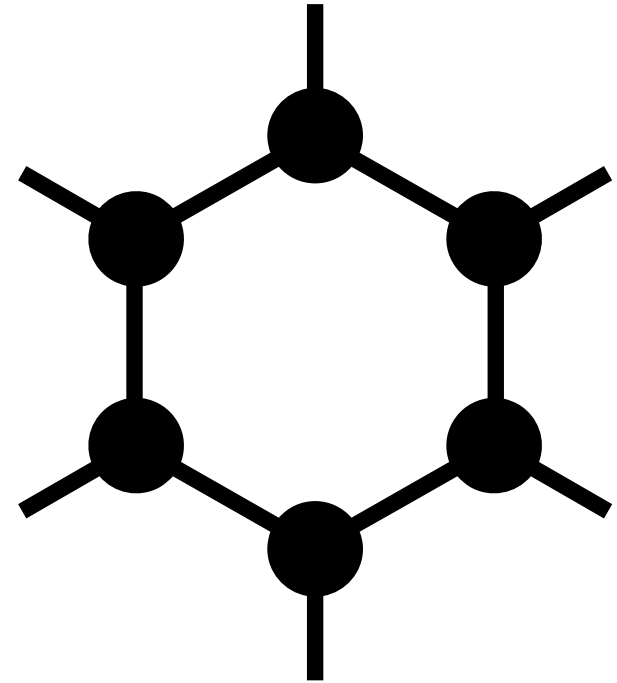
Murena /e/OS

- User friendly from the start
- Privacy-focused & deGoogled: uses MicroG, UnifiedPush, ...
- Can use Murena cloud with it
- Fork of LineageOS
- Supports 287 devices
- <https://e.foundation>



GrapheneOS

- Strong emphasis on [security hardening](https://grapheneos.org)
- Always ships latest security patches
- Only supports Google Pixel phones
- <https://grapheneos.org>



Magisk

- Manage modules on rooted devices
- E.g. useful for full backups
- <https://github.com/topjohnwu/Magisk>

Unlocking the bootloader

- Necessary to flash custom ROMs
- Process varies depending on device
- e.g. check [this list](#)
- Will erase all device data

Live install

<https://wiki.lineageos.org/devices/>
<https://lineage-archive.timschumi.net/>

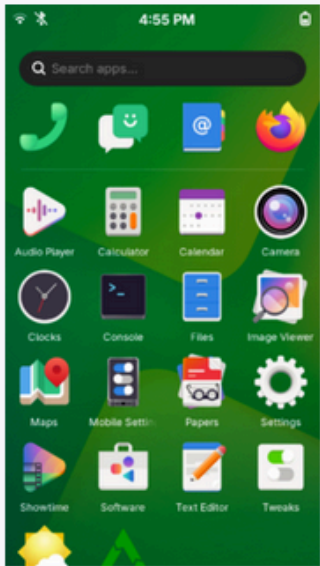
Linux phones

postmarketOS

- Linux on ARM → Alpine Linux spin-off
- By now usable Android alternative
- <https://postmarketos.org/>

Supports many interfaces

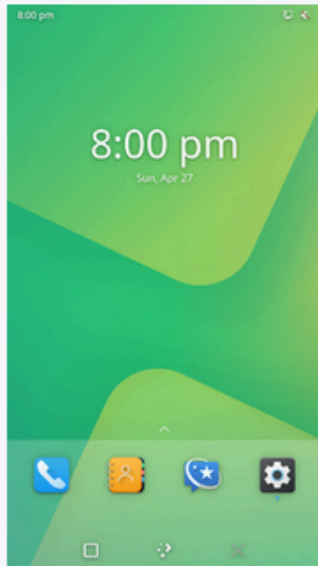
Phosh



A mobile interface originally developed by Purism. It uses the GNOME stack.

[Wiki page](#)

Plasma Mobile



The mobile version of KDE Plasma. It is based on the KDE stack.

[Wiki page](#)

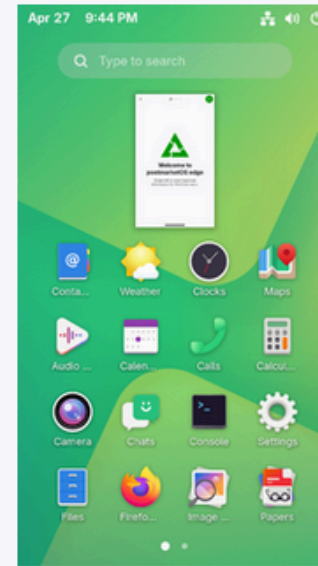
Sxmo



Based on a tiling window manager and uses hardware buttons as controls. For advanced users.

[Wiki page](#)

GNOME Mobile



The GNOME stack adapted for handsets.

[Wiki page](#)

Lomiri



An elegant and convergent shell for both desktop and mobile.

[Wiki page](#)

SailfishOS

- Backed by Finnish company Jolla
- Proprietary UI and apps :-/
- <https://sailfishos.org/>

Running Android apps

- WayDroid
- Android Translation Layer

Notifications

- Emulating MicroG can provide Firebase Cloud Messaging
- But continuous emulation is heavy on the battery!

Conclusion

<https://keepandroidopen.org/>

106

DAYS UNTIL
LOCKDOWN

Starting September 2026, a silent update, nonconsensually pushed by Google, will **block every Android app** whose developer hasn't registered with Google, signed their contract, paid up, and handed over government ID.

Every app and every device, worldwide, with no opt-out.



SPREAD THE WORD

✕ Post on X

🐙 Post on Mastodon

✂ Post on Bluesky

🌐 LinkedIn

f Facebook

What Google is doing

In August 2025, Google [announced](#) a new requirement: starting September 2026, **every Android app developer must register centrally with Google** before their software can be installed on any device. Not just Play Store apps: all apps. This includes apps shared between friends, distributed through [F-Droid](#), built by hobbyists for personal use. Independent developers, church and community groups, and hobbyists alike will all be frozen out of being able to develop and distribute their software.

Registration requires:

🔑 Paying a fee to Google

📄 Agreeing to Google's Terms and Conditions

Other info

- List of FOSS apps: <https://github.com/offa/android-foss>
- Check app compatibility on de-Googled phones: <https://plexus.techlore.tech/>

Feedback: feedback@feedback.thealternative.ch